

The built-in chains for the filter table are as follows.

Input chain: The input chain is applied to network packets that are targeted at the host. It is responsible for handling incoming connections to the firewall. If a user tries to SSH into any server, Iptables will attempt to match the IP address and port to a rule in the input chain.

Output chain: The output chain is used for outgoing connections, i.e., packets generated locally and going out from the local server.

Forward chain: This is applied to network packets routed through the host.

NAT table

The built-in chains for the NAT table are as follows.

- **Pre-routing:** Alters network packets when they arrive. This is used in Destination Network Address Translation.
- **Output:** Alters locally-generated network packets before they are sent out.
- **Post-routing:** Alters network packets before they are sent out.

Mangle table

The built-in chains for the Mangle table are as follows.

- **Input:** Alters network packets targeted for the host.
- **Output:** Alters locally-generated network packets before they are sent out.
- **Forward:** Alters network packets routed through the host.
- **Pre-routing:** Alters incoming network packets before they are routed.
- **Post-routing:** Alters network packets before they are sent out.

Raw table

- **Output:** Applies to locally-generated network packets before they are sent out.
- **Pre-routing:** Applies to incoming network packets before they are routed.

Security table

- **Input:** Applies to network packets targeted for the host.
- **Output:** Applies to locally-generated network packets before they are sent out.
- **Forward:** Applies to network packets routed through the host.

Every network packet received by or sent from a Linux system is subject to at least one table. However, a packet may be subjected to multiple rules within each table before emerging at the end of the chain. The structure and purpose of these rules may vary, but they usually seek to identify a packet coming from or going to a particular IP address, or set of addresses, when using a particular protocol and network service. Figure 1 outlines how the flow of packets is

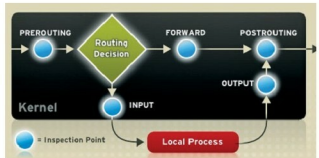


Figure 1: Packet filtering in Iptables

examined by the Iptables sub-system.

Iptables command options

The Iptables command options are as follows.

-A: Appends the rule to the end of the specified chain.

Unlike the **-I** option described below, it does not take an integer argument. It always appends the rule to the end of the specified chain.

-D <integer> | <rule>: Deletes a rule in a particular chain by number (such as 5 for the fifth rule in a chain), or by rule specification. The rule specification must exactly match an existing rule.

Example:

```
Iptables -D INPUT --dport 80 -j DROP, Iptables -D INPUT 1
```

This command deletes a rule in a chain.

-E: Renames a user-defined chain, which is any chain other than the default, pre-existing ones. (Refer to the **-N** option, below, for information on creating user-defined chains.) This is a cosmetic change and does not affect the structure of the table.

Example:

```
Iptables -E allowed disallowed
```

In this example, we changed the name of the chain from *allowed* to *disallowed*.

-F: Flushes the selected chain, which effectively deletes every rule in the chain. If no chain is specified, this command flushes every rule from every chain.

Example:

```
Iptables -F INPUT This command flushes all rules from the specified chain.
```

-h: Provides a list of command structures, as well as a quick summary of command parameters and options.

-I [<integer>]: Inserts the rule in the specified chain at a point specified by a user-defined integer argument. If no argument is specified, the rule is inserted at the top of the chain.